Food Remedy API – Capstone Gopher

RE006 – Historical Deployment and Recommendation Research Review

1. Summary

This report reviews the historical deployment and recommendation research completed during **Project A** of the **Gopher Industries Food Remedy API** project and identifies findings that remain relevant to the current **Project B** implementation. The historical research covered deployment readiness, cloud infrastructure, Firebase and Firestore integration, nutrition research, recommendation systems, mobile application deployment, App Store requirements, privacy, usability, and AI integration. This review evaluates which findings continue to support the current architecture based on **Firebase Authentication, Firestore, and Expo React Native**, removes outdated or duplicated recommendations, and maps the remaining findings to current deployment and recommendation pipeline decisions.

The Food Remedy API project has progressed from an MVP developed during Project A into a more mature application in Project B. During Project A, the research team investigated a wide range of topics that supported the technical and functional development of the application. These included cloud deployment, recommendation systems, nutrition science, user experience, privacy, accessibility, and production readiness.

The purpose of this review is to determine which historical research findings continue to provide value for the current system and how these findings influence deployment strategies and recommendation workflows in Project B.

2. Purpose and Scope

This report prevents the current team from re-researching ground already covered, and to stop outdated or duplicated conclusions from silently carrying forward into deployment or recommendation-pipeline decisions. The review covers prior research on:

- Product launch, app-store submission and mobile UX readiness.
- User retention, checkout behavior and cart design.
- Nutrition science, dietary restrictions and calorie/nutrient scoring.
- Recommendation-system design, personalization and explainability.
- Deployment, performance, monitoring, costing and privacy compliance.

Findings are evaluated against the architecture actually in use today – Firebase, Firestore and Expo – rather than against the general subject matter alone.

3. Review Methodology

- Each historical research item was read against its stated topic and original author's focus area.

- Items were checked for direct relevance to a current, named part of the system: the Firebase/Firestore backend, the Expo mobile client, the scrape → clean → enrich → seed pipeline, or the recommendation/checkout workflows.
- Items covering the same subject from more than one contributor were consolidated and logged once, under "duplicated."
- Items tied to features not yet built (e.g. AI/LLM recommendations, wearable integration) were flagged as forward-looking or out of current scope rather than discarded outright, unless no roadmap link exists.
- No historical recommendation was found to actively contradict another; overlaps were treated as duplication rather than conflict.

4. Findings Retained for Current Use

The following findings remain valid and are mapped to a specific current deployment or pipeline decision. They should be treated as reusable inputs rather than re-researched.

Historical Finding	Original Focus	Status	Current Mapping (Firebase / Firestore / Expo)
Dietary restriction & allergy classification (vegan, vegetarian, Jain, Halal, Kosher; nutrient deficiencies; restricted-ingredient handling).	Nutrition science research (multiple contributors)	Retained	Directly supports the allergen-detection and dietary-tag logic in the Database team's scrape → clean → enrich → seed pipeline, and the allergen/dietary fields already surfaced on the product-detail screen.
Calorie density, macronutrient composition, and nutrition-scoring/label research.	Core nutrition research	Retained	Underpins the calorie and nutrient-tracking fields used by the meal-logging feature and the nutrition-scoring logic in the enrichment pipeline.
Personalization using user goals, dietary preferences and demographic/profile data.	Recommendation-system research	Retained	Matches the current profile/demographic linkage used to drive the personalized recommendation workflow.
Context-aware recommendation inputs (time, location, activity, real-time signals).	Recommendation-system research	Retained, forward-looking	Applicable once the Recommendation page moves off its current hard-coded state; earmarked for the AI/LLM recommendation expansion rather than the present MVP build.
Real-time allergen warnings and dynamic in-app feedback during browsing and checkout.	UX / recommendation research	Retained	Supports the existing allergen-alert behavior on product-detail and cart screens; should extend into any future recommendation-engine rebuild.

Historical Finding	Original Focus	Status	Current Mapping (Firebase / Firestore / Expo)
Checkout-flow friction reduction, trust signals and payment UX; cart layouts and item-update patterns.	Checkout & cart UX research	Retained	Consistent with the implemented Shopping Cart and Checkout workflows; useful reference when the Compare/Recommendation cart logic is rebuilt.
App Store / Play Store submission requirements: metadata, privacy policy, review process, rejection risks.	Launch readiness research	Retained	Directly relevant to Expo/EAS build and submission; the current build is Android-focused, so this research should now be re-checked against iOS submission requirements before an iOS release.
Mobile UX readiness benchmarks against iOS/Android platform guidelines.	Launch readiness research	Retained	Feeds directly into the outstanding cross-platform (iOS) deployment gap flagged in the current status report.
Mobile performance optimization: load times, caching, lazy loading, API/query optimization.	Deployment research	Retained	Applies to Firebase ↔ SQLite offline-sync performance and to easing the Firebase read/write quota risk already identified as a scalability concern.
Privacy compliance: GDPR, user consent, data transparency, ethical handling of user data.	Compliance research	Retained	Feeds into Firestore security rules and the still-outstanding Firebase App Check / deployment-hardening work.
Post-deployment monitoring, analytics platforms and user tracking.	Deployment research	Retained	Matches the current priority to strengthen deployment monitoring and end-to-end testing before production release.
Evaluation metrics for recommendation quality: engagement, conversion, satisfaction.	Recommendation-system research	Retained, forward-looking	Should be applied once the recommendation engine is rebuilt on real logic rather than the current placeholder implementation.
Deployment and operational costing: hosting, backend infrastructure, scalability, maintenance.	Deployment research	Retained	Directly relevant to the identified risk that Firebase alone may not scale, and to the recommendation to plan a dedicated backend API layer.

5. Findings Removed or Deprioritized

The following findings were removed from active consideration, or deprioritized until a later stage of the roadmap, for the reasons stated.

Historical Finding	Original Focus	Status	Current Mapping (Firebase / Firestore / Expo)
Basic definitions of what calories are and how they are reported on labels.	Foundational nutrition research (multiple contributors)	Removed – duplicated / settled	Definitional groundwork only; superseded by the calorie/nutrient tracking already implemented. Kept as background reference, not an active input.
Hormonal influence on metabolism, appetite and nutritional needs across life stages (independently researched three times).	Nutrition science research	Removed – duplicated	Three overlapping research threads on the same subject; no unique implication for the Firebase/Firestore data model beyond what is already captured in the profile and dietary-need fields. Consolidated out.
Cross-platform consistency of personalized nutrition systems across mobile, web and integrated services.	Personalization research	Removed – out of current scope	The current build targets a single Expo/React Native mobile app with no web client in scope, so this finding does not map to an active decision. Revisit only if a web client is added to the roadmap.
Generic mobile app testing methodology overview (unit, integration, usability, performance testing).	Testing research	Removed – superseded	Superseded by the end-to-end, synchronization-behavior and deployment-readiness testing already carried out in T1 2026. Retained only as a general checklist, not a pipeline input.
Explainable-recommendation transparency (labels, scores, descriptive insights) as a near-term feature.	Recommendation-system research	Deprioritized – premature	The Recommendation and Compare features are currently incomplete or hard-coded, so explainability work has nothing to attach to yet. Held for the AI/LLM recommendation rebuild rather than immediate deployment decisions.
Wearable-device integration and advanced health-data interoperability.	Planned future features (prior team)	Removed – not on current roadmap	No corresponding Firebase/Firestore schema or Expo module work exists for this. Out of scope until a future team formally adopts it.

6. Mapping to Current Deployment and Pipeline Decisions

6.1 Immediately actionable

- **Compliance & security –** prior GDPR/consent/privacy research should directly inform closing the outstanding Firebase App Check and deployment-hardening gap.
- **iOS submission –** prior App Store/Play Store and platform-UX research should be re-checked now, since the current build is Android-focused and iOS deployment preparation is still incomplete.

- **Performance –** prior caching/lazy-loading/API-optimization research applies directly to reducing Firebase read/write volume, which is the team's current top scalability risk.
- **Data pipeline –** prior dietary-restriction and allergen research should continue to govern the field structure used in the clean → enrich → seed pipeline; no changes needed, only continued application.

6.2 Hold until the recommendation engine is rebuilt

- **Context-aware and explainable recommendations –** these findings are sound but currently have nothing to attach to, since the Recommendation and Compare pages are still hard-coded or incomplete. Apply once that logic is rebuilt, not before.
- **Recommendation evaluation metrics –** engagement/conversion/satisfaction metrics research should be revisited once real recommendation output exists to measure.

6.3 Not currently in scope

- **Cross-platform (web) consistency –** no web client exists in the current roadmap; this research stays archived until those changes.
- **Wearable-device integration –** no corresponding schema or module work exists yet; treat as a future-team item only.

7. Conclusion

The historical research completed during Project A continues to provide a strong foundation for the Food Remedy API in Project B. Research relating to deployment, cloud infrastructure, recommendation systems, nutrition analysis, accessibility, privacy, and mobile application development remains directly relevant to the current Firebase, Firestore, and Expo architecture.

Several early planned activities, including initial Firebase evaluation and deployment preparation, have now been completed and therefore no longer require additional investigation. However, the majority of the historical research continues to guide current deployment decisions and recommendation pipeline development.

Overall, Project B benefits from building upon the validated findings established during Project A rather than repeating previous research. This allows the project to focus on improving recommendation quality, strengthening deployment readiness, enhancing user experience, and preparing the Food Remedy API for future production deployment and AI-supported functionality.

8. Reference

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